



PAWAN KUMAR

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Objective

Motivated Computer Engineering student at Swarnnim University, proficient in Python, Java, and Full-Stack Development, with a strong interest in AI/ML and Software Engineering. Seeking an internship/role to apply my problem-solving skills and technical knowledge to develop innovative solutions while advancing my expertise in emerging technologies.

Education

Course / Degree	School / University	CGPA	Year
B.Tech in Computer Engineering	Swarnnim Startup & Innovation University	9.02	Currently Pursing
Higher Secondary School	Hetauda School of Management	5.8	2021
Secondary School	Shree Adhunik Rastriya Higher Secondary School	7.1	2019

Projects

- **EmpowerAbility: Mapping the Missing Limbs [March'25]**

Technologies: (Python, Streamlit, Mediapipe Holistic, YOLOv8, OpenCV)

Project Definition: Developed a real-time AI monitoring system in Python to enhance accessibility by detecting disabled individuals in public spaces. The system leverages OpenCV for live video processing and frame analysis, Streamlit for an interactive web dashboard, MediaPipe Holistic for human pose estimation to identify mobility patterns, and YOLOv8 for object detection of assistive devices like wheelchairs and crutches. This solution provides actionable insights for smart cities and healthcare facilities by identifying accessibility challenges in real-time, enabling timely assistance and infrastructure improvements.

My Role: Served as project leader, overseeing end-to-end development while actively contributing to core coding task.

- **Photo Enhancer: AI-powered Photo Enhancement [April'25 – April'25]**

Technologies: (Python, Streamlit, OpenCV, GFPGAN, RealESRGAN)

Project Definition: Built an intelligent image restoration system that automatically enhances blurry, low-resolution, or damaged photographs. Leveraged OpenCV for core image processing, GFPGAN for facial reconstruction, and RealESRGAN for 4x super-resolution upscaling, deployed via a Streamlit web interface. The tool effectively restores archival photos, improves forensic evidence quality, and assists digital artists by recovering lost details while preserving natural textures.

My Role: As a sole Developer and Researcher, I undertook full stack responsibility for the project, including system architecture design, algorithm development, implementation, testing, optimization and deployment.

- **Crook Catcher: AI-Powered Face Identification [March'25 – March'25]**

Technologies: (Python, OpenCV, Face Recognition, MobileNet SSD, NumPy)

Project Definition: Developed a real-time person identification system that classifies known and unknown individuals using face recognition. Implemented ROI selection for targeted monitoring, MobileNet SSD for human detection, and face encoding/recognition to match against a database. The system automatically captures and organizes images of detected persons with timestamps, providing valuable data for security applications.

My Role: As a sole Developer and Researcher, I undertook full stack responsibility for the project, including system architecture design, algorithm development, implementation, testing, optimization and deployment.

Technical Skills

- Python
- Java
- JavaScript
- HTML, CSS
- C

Non – Technical Skills

- Communication Skills
- Time Management
- Integrity

Achievements

- Finalist, *Code Unnati Innovation Marathon 3.0* (2025) – AI-based solution for social impact.
- Grant Recipient (₹2 Lakhs), Gujarat Student Startup & Innovation Policy (SSIP) for *EmpowerAbility* (2025)

Certifications

- SAP Certification in Python for Machine Learning & Deep Learning [July'23 - August'23]

Industrial Exposer

Adani Group Industrial Visits [March'25]

(Adani Port & SEZ | Adani Wilmar Ltd | Solar Plant | Windtech)

- Studied port automation systems, edible oil production IoT, and renewable energy infrastructure
- Analyzed solar/wind power generation workflows and sustainability practices
- Documented industrial safety protocols and large-scale supply chain management.

Indian Space Research Organisation (ISRO), Ahmedabad [May'23]

- Observed satellite assembly, testing facilities, and space technology R&D at SAC (Space Applications Centre)
- Gained insights into remote sensing, launch vehicle systems, and mission control operations
- Interacted with scientists about software systems for spacecraft navigation and data processing

Co-curricular Activities

- 1st Prize, Intercollege Football Tournament [January'25]
- Passionate multi-instrumentalist (guitar, piano, drums) fostering creativity, discipline, and collaboration through music.

Personal Details

- Nationality: Nepali
- Date Of Birth: October 12, 2003
- Languages Known: English, Hindi, Nepali, and Bhojpur
- Interests: Chess and Travelling
- Address: Swarnim Hostel, Bhoyan Rathod, Gandhinagar, 382721